

Recovering a denoiser's implicit regularizer: three routes

u_{PM} is the prox of an implicit regularizer (Gribonval), so its optimality condition pins any correct prior on $\text{range}(u_{\text{PM}})$ — and no ground truth is needed:

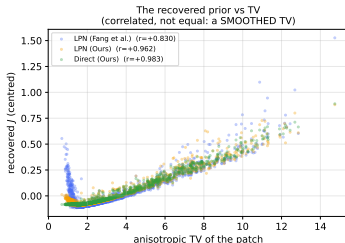
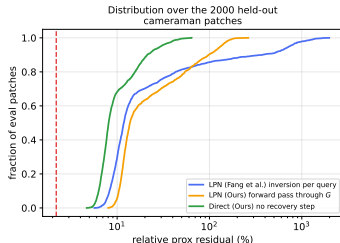
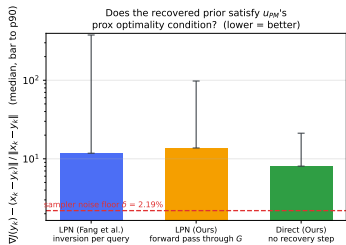
$$u_{\text{PM}}(x) = \text{prox}_{f_{\text{reg}}}(x) \iff \nabla f_{\text{reg}}(y_k) = x_k - y_k, \quad y_k = u_{\text{PM}}(x_k), \quad \text{read against the noise floor } \delta.$$

Iterative	$J_1 = \langle x, w \rangle - \psi_\theta(w) - \frac{1}{2} \ x\ ^2, \quad \nabla \psi_\theta(w) = x$ (inversion per query)
One-shot	$J_2 = G_\theta - \frac{1}{2} \ x\ ^2, \quad G_\theta \approx \psi_\theta^*$ (one forward pass)
Direct fit	J_θ fitted directly, $\nabla J_\theta(y_k) \approx x_k - y_k$ (no recovery step)

J_1, J_2 are 1-semiconvex for free — the $-\frac{1}{2} \|\cdot\|^2$ is in the Fenchel formula. J_θ is a plain ICNN, so it *assumes* f_{reg} is convex: true for TV, false in general.

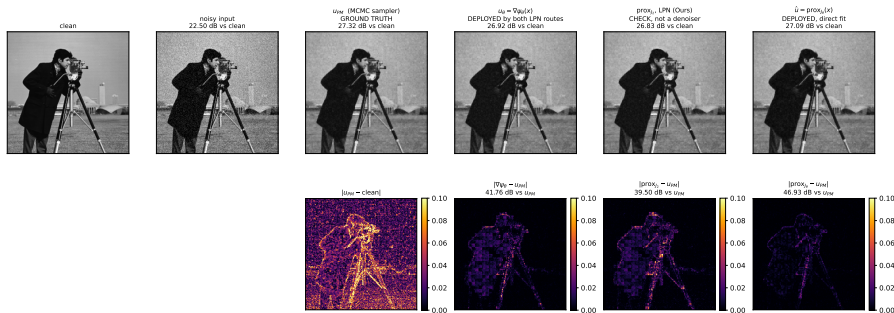
prior	resid	p90	$\times \delta$	cost
Iterative	14.48%	371%	6.6	12.7 s
One-shot	13.73%	98%	6.3	0.03 s
Direct fit	8.07%	21%	3.7	0.03 s

2000 held-out cameraman patches, $\delta = 2.19\%$. $\nabla \psi_\theta$ vs u_{PM} : 0.41% median — everything above that is recovery-step error.



Iterative has *no safe* α : at $\alpha = 0$ the solve converges but 0.45% of patches leave the valid box; at $\alpha = 0.1$ the $O(\alpha)$ bias takes it to 99.55%.

The prior in action, and what the comparison establishes



method	PSNR	vs u_{PM}	rel- L^2
noisy input	22.50	—	—
u_{PM} (MCMC)	27.32	—	—
$u_\theta = \nabla\psi_\theta$	26.92	41.76	1.56%
prox_{J_2} (check)	26.83	39.50	2.02%
$\hat{u} = \text{prox}_{J_\theta}$	27.09	46.93	0.86%

Neither LPN route inverts anything to *denoise*: since $\nabla G = (\nabla\psi)^{-1}$, **Iterative** and **One-shot** both deploy $\nabla\psi_\theta$; they differ only in how they *evaluate* the prior. The prox_{J_2} panel is a consistency check, not a denoiser — for **Direct fit** that object *is* the deployed denoiser, so no gap can open.

1 — the class is not negotiable (mathematical). **Direct fit** assumes f_{reg} convex; on a nonconvex prior it reads 101% with curvature pinned at 0.000 against an exact -0.911 . No budget or dimension fixes it.

2 — among valid routes, the ranking is a solver question (numerical). Exact inversion (1-D): **Iterative** 0.02% vs **One-shot** 1.22%; Adam inversion at $d = 64$: **Iterative** trails, at $\sim 400\times$ the query cost.

Cameraman, $\sigma = t = 20/256$, $m = 8000$, all nets 250 000 steps. Reproduce: three_way_retrain.ipynb, PRESET="full".